

SOP and POS

Sum-of-products ORs minterms where F is one

Minterms and maxterms

- A minterm is a product true on exactly one row, like $A'B$ for m zero on two variables
- A maxterm is a sum false on exactly one row, like $A+B$ for M zero
- Canonical SOP lists every one row; canonical POS lists every zero row
- That full form is not always minimal, you still simplify with grouping or laws

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / SOP ↔ POS

INTERACTIVE TOOL

SOP ↔ POS converter

Fill one truth table and read both **canonical SOP** (minterms) and **POS** (maxterms) — dual forms of the same function.

Starter example: 2-variable XOR — $F=1$ on rows 1 and 2. SOP is $A'B + AB'$; POS uses the zero rows' maxterms.

Load starter example

Challenges 0 / 22 cleared

Quiz: SOP: SOP uses rows where F equals? Answer: 1

Answer

Show hint

Check

Next

Idle

Quiz: SOP

Quiz: POS

Quiz: minterm

Quiz: maxterm

Starter XOR

Derive XOR

XOR POS

Preset AND

Preset OR

All zero

All ones

Toggle F

Quiz: sigma

3 variables

Explain dual

Quiz: minimal?

AND POS

Quiz: m0

Quiz: M0

Build NAND

NAND POS

Full dual

Core ideas

SOP

Sum of products = OR of minterms where $F=1$.

POS

Product of sums = AND of maxterms where $F=0$.

SOP POS starter

learn_digital — sop pos

Workbook practice

- In the workbook track, take two-variable XOR
- Write canonical SOP from rows one and two, and POS from rows zero and three
- For AND, note only m three is one
- For all zeros, SOP is zero; for all ones, both forms collapse to one
- Name one pitfall: assuming canonical SOP is already minimal

Pitfalls to watch

- Do not mix SOP and POS rules, SOP uses F equals one, POS uses F equals zero
- Minterm m zero and maxterm M zero are different shapes
- And remember: the browser lab is literacy
- Real synthesis picks the form that fits your gate library and timing

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, write SOP and POS for one small table
- When you are ready, take the short quiz, then continue to don't-care minimization